

Passive Tone Detection for Moving Targets Based on Long-Time Coherent Integration

Liu Zhang , Shengchun Piao , Junyuan Guo , and Xiaohan Wang

Abstract—The detection of sinusoidal signals embedded in noise is an important topic in passive sonar signal processing. The performance of existing tone detection techniques is limited by the Doppler frequency shift, which is caused by the relative motion between moving targets and the receiver. In this study, an algorithm based on long-time coherent integration is proposed to achieve robust tonal signal detection under the influence of a Doppler frequency shift. First, the received narrowband signals radiated by moving targets are split into multiple segments through a window-length constraint. The results obtained by applying a discrete Fourier transform (DFT) to the data segments are modeled as a polynomial phase signal in the frequency domain. Then, the polynomial Radon-polynomial Fourier transform (PRPFT) is applied to simultaneously compensate the time-variant frequency drift and phase difference. To avoid the gain loss due to the discretization of the phase difference compensation in PRPFT, a phase compensation factor searching algorithm is proposed. The simulation results show that the proposed algorithm can provide higher frequency resolution and higher coherent integration gain even in the case of a severe Doppler shift. Furthermore, the results of sea trials demonstrate that the proposed method can perfectly achieve coherent integration for signals with a duration of several hundred seconds under different experimental conditions.

Index Terms—Long-time coherent integration, moving target, passive tone detection.

I. INTRODUCTION

NARROWBAND signal detection is an essential task in signal processing for passive sonar applications. Although the radiated noise generated by underwater targets consists of tonal components and broadband spectra [1], the tones exhibit better stability and higher power per unit bandwidth compared with other radiated components [2], [3]. For example, Burenkov et al. [4] have reported that the phase of tones is highly stable

even for signals propagated up to 9000 km. Therefore, tones are an important feature for passive sonar detection.

Power spectrum estimation (PSE) methods [5] are effective for detecting narrowband tones embedded in a noisy background. The periodogram [6] and Welch's methods [7], which are based on the fast Fourier transform (FFT), are two widely known classical spectral estimation methods. However, they often suffer from insufficient resolution. Modern spectral estimators can produce more accurate spectral estimates because the spectral estimate is derived from an assumed model that is a close approximation to reality [8], [9], [10], [11], [12], [13], such as the spectral estimation methods based on the autoregressive (AR), moving-average (MA), and autoregressive moving-average (ARMA) models. However, the performance advantages of modern spectral estimators strongly depend on the signal-to-noise ratio (SNR). In fact, modern spectral estimators are often worse than those obtained with conventional Fourier analysis [14], [15] in sufficiently low SNR situation. This situation makes the classical spectral estimators more attractive for real-time implementation.

A number of improved detectors have been developed on the basis of classical spectral detectors. Compared with the periodogram, Welch's Method (in which the periodograms for overlapping blocks are averaged linearly) can reduce the variance and sensitivity toward frequency fluctuations of the spectral estimates. The average power processor (AVGPR) [16], equivalent to Welch's method, was proposed to achieve the autonomous detection of narrowband signals. Later, Wagstaff et al. [16], [17], [18], [19] proposed the Wagstaff's integration silencing processor (WISPR) filter to perform nonlinear integration over the segmented periodograms. However, these methods are extremely sensitive to the low power at the signal frequencies. To circumvent this problem, an optimal tonal detector [20] was developed to obtain a more robust detection performance by utilizing the probability distribution of the segmented power spectrum.

The abovementioned processors achieve promising performances for tone detection. However, the gain of the noncoherent average detector will be limited if the SNR decreases. This degradation may be mitigated when the phase information of the tones is used.

Coherent detectors have been investigated by several researchers. Wang and Wan [21] designed a tonal detector that estimates and compensates the phase between segments to obtain coherent integration gain. Further, Chen [22] improved the accuracy of the phase estimation by applying three-point

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interpolation in the frequency domain. For these two detectors, the phase difference must be estimated first before realizing coherent integration. The complex estimation program will degrade the performance of these detectors when detecting weak targets. Recently, Lan et al. [23] proposed a flexible method for weak signal detection, i.e., the coherently averaged power spectral estimate (CAPSE) method. FFT technology was used to directly compensate the phase, and the procedure of parameter estimation was omitted. This greatly reduces the complexity of the coherent integration.

It is worth noting that all the aforementioned algorithms are based on the assumption that the target is stationary. In the case of a mobile target, the observed frequency of the received acoustic signals changes with time due to the Doppler effect [24], [25]. This may be the most serious deterrent for utilizing coherent signal processors over exceptionally long integration intervals [34]. In [26], a coherent detector utilizing the information of adjacent frequency bins was designed to gather the signal energy dispersed due to the frequency shift. However, when the frequency shift is extremely severe, the power is dispersed into several or even dozens of discrete frequency bins. To solve this problem, Guo et al. [27] invented a coherent integration algorithm based on the cross-power spectrum with the frequency shift and phase change corrected by a compensating time-scale (Doppler) factor and time delay. This algorithm can provide sufficient coherent integration gains for the detection of moving targets. But the estimation accuracy of the Doppler factor and time delay is extremely sensitive to the reduction of SNR, which may greatly reduce the gain of the algorithm in a strong noise environment.

Recently, Wu et al. [28] proposed a method combining the polynomial Radon and polynomial Fourier transforms (referred to as PRPFT) to detect maneuvering radar targets. The polynomial Radon transform was used to search for an arbitrary curve in the range–slow time plane, and simultaneously the polynomial Fourier transform was used to compensate the Doppler migration (i.e., the phase of the discrete points on the searched curve). This algorithm can effectively realize coherent integration for radar target echoes on the range–slow time plane [28], [29], [30]. However, the multiresolution searching strategy described in [28] suffers from gain loss due to the discretization of phase difference compensation.

In this study, we determine a tone detection method based on long-time coherent integration, that can achieve robust detection performance under the influence of the Doppler effect. The main contributions of this study include the following three aspects.

- 1) First, the model of signals received by a single hydrophone from a uniformly moving linear target in a passive system is established. By selecting an appropriate window length and overlap ratio, the DFT results of the signal segments are modeled as a polynomial-phase signal.
- 2) Second, to avoid the gain loss due to the discretization of the phase difference compensation in PRPFT, a phase compensation factor searching algorithm is proposed. Furthermore, a projection method according to the maximum energy point is proposed to maximize the ratio of time-variant tones relative to the level of background noise.

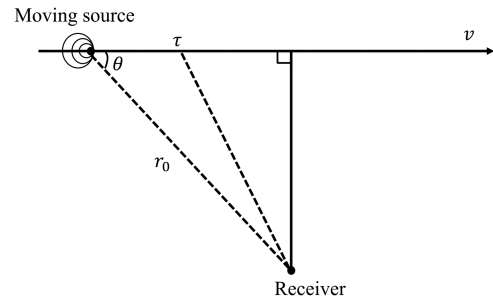


Fig. 1. Source-receiver geometry.

- 3) Finally, the effect of the overlapping segments on the coherent integration gain is analyzed via theoretical derivation to optimize the design of the overlap ratio. Compared with other methods, the proposed algorithm can provide a higher gain for the same coherent integration time. Besides, the maximum coherent integration time of the proposed algorithm is the longest of all evaluated methods under identical experiment conditions.

The rest of this article is organized as follows. Section II describes the proposed long-time coherent integrated line spectrum detection (LTCILSD) method, including the principle and the detailed analysis of integration gain. Numerical simulations and experimental data processing are discussed in Sections III and IV, respectively. Finally, Section V concludes this article.

II. LONG-TIME COHERENT INTEGRATED LINE SPECTRUM DETECTION

A. Modeling the Received Signals Radiated From a Moving Target

The relative motion between the target and receiver produces a change in the observed frequency of the received acoustic signals. We establish a model for this system as follows.

Consider a target moving along a straight line at a constant speed v through a stationary receiver, as described in Fig. 1. The initial distance between the source and receiver is r_0 . At that instant, the intersection angle between the source-receiver line and the motion in the source direction is denoted by θ . A pure tone of frequency f_0 emitted by the source at a time τ arrives at the receiver at a later time t . According to [31], τ is given by

$$\tau = \frac{c^2 t - \Lambda - v r_0 \cos(\theta)}{c^2 - v^2} \quad (1)$$

where

$$\Lambda = \sqrt{v^2 c^2 t^2 - 2 v r_0 c^2 \cos(\theta) t + r_0^2 [v^2 \cos^2(\theta) + c^2 - v^2]}$$

and c is the propagation velocity of sound in water.

As a result, the theoretical expression of the received signal can be represented as

$$x(t) = A \exp \left\{ j 2 \pi f_0 \frac{c^2 t - \Lambda - v r_0 \cos(\theta)}{c^2 - v^2} \right\} \quad (2)$$

where A is the amplitude of the received signal. According to the Weierstrass Approximation Theorem, (2) can be approximated

as a polynomial phase signal [32] of order Υ with the following expression:

$$x(t) = A \exp \left\{ j2\pi \sum_{\lambda=0}^{\Upsilon} \mu_{\lambda} t^{\lambda} \right\}, \quad t \in [0, T] \quad (3)$$

where $\mu_{\lambda}, \lambda = 0, 1, \dots, \Upsilon$ are real coefficients, and T is the signal length. The model expressed in (3) indicates that the received signal is nonstationary and time-varying due to the Doppler effect. In other words, the instantaneous frequency of the received signal varies nonlinearly. This phenomenon can be further analyzed in the frequency domain, as described in the following section.

B. Observations in the Frequency Domain

High coherent integration gain can be obtained by performing long-time DFT on a signal. However, as the processing data length increases, the signal power will be dispersed into multiple frequencies due to the Doppler shift. Under this condition, the source frequency cannot be extracted correctly, and the integration gain will not increase accordingly. In order to reduce the sensitivity of frequency resolution cell to the Doppler shift, segmentation processing must first be utilized.

Assume that the received signal in (3) is sampled at time interval T_s . The available signal of N samples may be split into segments of M samples each, which are partially overlapped by $M - D$. Then, the signal in the k^{th} segment is given by

$$x_k(n) = Aw(n + kD) \exp \left\{ j2\pi \sum_{\lambda=0}^{\Upsilon} \mu_{\lambda} (nT_s)^{\lambda} \right\} \\ n = 0, 1, \dots, M - 1, k = 0, 1, 2, \dots, K - 1 \quad (4)$$

where $w(n), n = 0, 1, \dots, M - 1$ is a time domain window function. For convenience, a rectangular window is used in the following discussion. K represents the total number of data segments.

After applying a DFT to each data segment, we can obtain

$$X(f_l, k) \\ = \frac{1}{M} \sum_{n=0}^{M-1} A \exp \left\{ j2\pi \sum_{\lambda=0}^{\Upsilon} \mu_{\lambda} [(n + kD)T_s]^{\lambda} \right\} \\ \exp \{-j2\pi f_l T_s n\} \\ = \frac{A}{M} \sum_{n=0}^{M-1} \exp \left\{ j2\pi \sum_{\lambda=0}^{\Upsilon} \mu_{\lambda} (kDT_s)^{\lambda} \right\} \\ \times \exp \left\{ j2\pi \left[\left(\sum_{\lambda=1}^{\Upsilon} \lambda T_s \mu_{\lambda} (kDT_s)^{\lambda-1} - f_l T_s \right) n + O_n \right] \right\} \quad (5)$$

where $f_l = l/MT_s, l = 0, 1, \dots, M - 1$, l represents the discrete frequency index and O_n represents the sum of higher powers (more than the first power) of n .

Notice that the frequency change in each segment is $|\sum_{\lambda=2}^{\Upsilon} \lambda \mu_{\lambda} (MT_s)^{\lambda-1}|$, and the frequency resolution is

$1/MT_s$ Hz. A limitation is imposed where the frequency variation of each segment does not exceed a frequency resolution cell. That is

$$\left| \sum_{\lambda=2}^{\Upsilon} \lambda \mu_{\lambda} (MT_s)^{\lambda-1} \right| \leq \frac{1}{MT_s}. \quad (6)$$

Under this condition, the signal frequency in each segment can be regarded as a constant. Here, we assume this constant is the initial frequency $\sum_{\lambda=1}^{\Upsilon} \lambda \mu_{\lambda} (kDT_s)^{\lambda-1}$ of each segment. Thus, each signal segment can be approximated as a complex exponential signal, and the term O_n in (5) can be omitted. Therefore, (5) can be approximated as

$$X(f_l, k) \approx A \frac{\text{sinc}(\pi MT_s F_k)}{\text{sinc}(\pi T_s F_k)} \exp \{j\Phi_M\} \exp \{j\Phi_S\} \quad (7)$$

where

$$F_k = \sum_{\lambda=1}^{\Upsilon} \lambda \mu_{\lambda} (kDT_s)^{\lambda-1} - f_l \quad (8)$$

$$\Phi_M = \pi T_s (M - 1) \left[\sum_{\lambda=1}^{\Upsilon} \lambda \mu_{\lambda} (kDT_s)^{\lambda-1} - f_l \right] \quad (9)$$

$$\Phi_S = 2\pi \sum_{\lambda=0}^{\Upsilon} \mu_{\lambda} (kDT_s)^{\lambda}. \quad (10)$$

The first exponential term in (7) is a linear term, which is related to the instantaneous frequency of the received signal, and the second exponential term in (7) is a polynomial representing the piecewise deformation caused by the unknown offset signal length D .

From (7), the peaks of the results after DFT are approximately distributed along a polynomial curve $f_l = \sum_{\lambda=1}^{\Upsilon} \lambda \mu_{\lambda} (kDT_s)^{\lambda-1}$. More importantly, the time series of the peaks is a polynomial phase signal with respect to segments k . Thus, the key of the coherent signal processors is to implement compensation for the polynomial frequency shift and polynomial phase difference. In the proposed design, the offsets are corrected simultaneously on the basis of the modified PRPFT.

C. Gain Loss of the Coherent Integration Using PRPFT

Generally, the definition of PRPFT for the complex function $X(f_l, k)$ referred in [28] is written as

$$G(\alpha_1, \alpha_2, \dots, \alpha_{\Upsilon}, \beta_1, \beta_2, \dots, \beta_{\Upsilon}) \\ = \sum_{k=0}^{K-1} X \left[\sum_{\lambda=1}^{\Upsilon} \lambda \alpha_{\lambda} (kDT_s)^{\lambda-1}, kDT_s \right] \\ \times \exp \left\{ -j2\pi \sum_{\lambda=1}^{\Upsilon} \beta_{\lambda} (kDT_s)^{\lambda} \right\} \quad (11)$$

where $\alpha_1, \alpha_2, \dots, \alpha_{\Upsilon}$ and $\beta_1, \beta_2, \dots, \beta_{\Upsilon}$ are discrete search parameters. The essence of PRPFT is to coherently accumulate the energy of the corresponding frequency for each moment kDT_s .

Theoretically, PRPFT can realize a long-time coherent integration for the moving target with arbitrary Doppler shifts.

However, multidimensional search for high-order PRPFT will produce heavy computational burden. In practice, the speed of an underwater moving target is considerably less than the sound speed in water. Serious Doppler shift exists only when the target is around the closest points of arrival. By experiments, it is found that as long as $v^2(t - (r_0 \cos(\theta)/v))^2 \leq 0.7r_0^2 \sin^2(\theta)$ is satisfied, the change of instantaneous Doppler shift can be approximated as linear (see Appendix I for details). Therefore, the effect of the third-order and above can be ignored. Equations (6) and (8)–(11) can be simplified as

$$M \leq \frac{1}{\sqrt{|2\mu_2 T_s^2|}} \quad (12)$$

$$F_k = \mu_1 + 2\mu_2 kDT_s - f_l \quad (13)$$

$$\Phi_M = \pi T_s (M - 1)(\mu_1 + 2\mu_2 kDT_s - f_l) \quad (14)$$

$$\Phi_S = 2\pi \left[\mu_0 + \mu_1 kDT_s + \mu_2 (kDT_s)^2 \right] \quad (15)$$

$$G(\alpha_1, \alpha_2, \beta_1) = \sum_{k=0}^{K-1} X(\alpha_1 + \alpha_2 kDT_s, kDT_s) \times \exp \left\{ -j2\pi \left[\beta_1 kDT_s + \frac{1}{2} \alpha_2 (kDT_s)^2 \right] \right\} \quad (16)$$

where $2\pi\mu_0$ is the initial phase of the received signal. μ_1 and $2\mu_2$ represent the initial frequency and chirp rate of the received signal, respectively. α_1 , α_2 , and β_1 are discrete search parameters.

Equation (14) can be viewed as the combination of the Radon transform and the second-order polynomial Fourier transform. The across-frequency-bins frequency shift in (13) and linear phase difference in (14) are corrected when the searching line of the Radon transform is matched to the frequency shift. Simultaneously, the polynomial Fourier transform [28], [38] is used to compensate the polynomial phase difference in (15). Therefore, when α_1 , α_2 , and β_1 are matched to μ_1 , $2\mu_2$, and μ_1^* (where μ_1^* represents the equivalent initial frequency, which will be defined in the following), the energy of the received signal can be accumulated at a coherent peak in the parameter domain $G(\alpha_1, \alpha_2, \beta_1)$ coherently.

The first-order phase in (15) is $2\pi\mu_1 kDT_s$. The digital frequency l_0^* corresponding to μ_1 is $l_0^* = \mu_1 MT_s$, which can be described as $l_0^* = l_0 + \delta$. Where l_0 is an integer that is the closest to l_0^* and δ is a decimal fraction. A multiresolution searching strategy was designed in [28] to implement PRPFT. It utilizes FFT to compensate the first-order phase, which can be given as

$$G_p = \max_{l'} \left| \frac{1}{K} \sum_{k=0}^{K-1} e^{j2\pi\mu_1 kDT_s} e^{-j2\pi l' k/K} \right|^2 = \frac{1}{K^2} \max_{l'} \left| \frac{\sin[\pi(\delta K - l')]}{\sin[\pi(\delta K - l')/K]} \right|^2. \quad (17)$$

The relationship $D = M$ is assumed in (17). l' is the discrete frequency index. When $l' = \delta K$, the first-order phase $2\pi\mu_1 kDT_s$ is exactly compensated by (17). But the discrete l' is not equal to δK in most cases. The maximum of $|\delta K - l'|$

is 0.5. So the maximum gain loss is

$$10 \log_{10} \left[\frac{1}{\left| \frac{1}{K^2} \left| \frac{\sin[\pi/2]}{\sin[\pi/2K]} \right|^2 \right|} \right]$$

which increases with the increase of K . When $K \geq 10$, the maximum gain loss is approximately constant at 3.9 dB. This loss may be very important for detecting signal with Doppler shifts and it can be avoided by more accurate compensation of the phase difference.

Besides, the spectrum estimation results were obtained by taking the maximum of the FFT across segments in a multiresolution searching strategy. The large energy components of interest are retained using this strategy. However, the act of taking the maximum in the frequency bins where there is no signal of interest can inevitably lead to gain loss, which will be shown in Section II-F.

D. Coherent Integration Based on the Modified PRPFT

Another implementation of PRPFT, a phase compensation factor searching algorithm, is proposed to improve the compensation accuracy of the phase difference. In addition, the projection method according to the maximum energy point is designed to reduce the gain loss caused by the process of taking maximum in PRPFT.

1) *Searching Parameter Space:* Given that the value of F_k in (13) is zero when the searching parameters α_1 and α_2 , respectively, match the initial frequency μ_1 and chirp rate $2\mu_2$ of the signal, then by substituting (11) and (13)–(15) into (16), the first-order phase is extracted and is expressed as

$$\phi_k = 2\pi kDT_s(\mu_1 - \beta_1). \quad (18)$$

In (18), μ_1 is unknown but is a constant, and D is known. It is well known that the period of $\exp\{j\phi_k\}$ is 2π , i.e.,

$$\exp\{j2\pi kDT_s(\mu_1 - \beta_1)\} = \exp\{j2\pi kDT_s(\mu_1^* - \beta_1)\}, \quad (19)$$

where $0 \leq \mu_1^* \leq 1/DT_s$ is defined as the equivalent initial frequency. It could now be argued that in order to compensate the first-order phase difference between adjacent segments without the influence of a periodic sidelobe, the following requirement should be fulfilled:

$$0 \leq \beta_1 \leq \frac{1}{DT_s}. \quad (20)$$

From the preceding analysis, the searching parameters with the searching variables i, j, m in (16) can be defined as

$$\alpha_{1i} = \alpha_{1\min} + \frac{i}{MT_s}, i = 0, 1, \dots, N_{\alpha_1} - 1 \quad (21)$$

$$\alpha_{2j} = \alpha_{2\min} + j\Delta\alpha_2, j = 0, 1, \dots, N_{\alpha_2} - 1 \quad (22)$$

$$\beta_{1m} = m\Delta\beta_1, m = 0, 1, \dots, N_{\beta_1} - 1 \quad (23)$$

where N_{α_1} , N_{α_2} , and N_{β_1} are the numbers of search parameters α_1 , α_2 , and β_1 . $\Delta\alpha_2$ and $\Delta\beta_1$ are the search intervals of α_2 and β_1 .

It is necessary to emphasize that the compensation accuracy of the frequency shift and phase difference increases with the

decrease in $\Delta\alpha_2$ and $\Delta\beta_1$, respectively. However, the effect of the searching intervals must be carefully examined because the improvement in compensation accuracy slows down and the computational load increases rapidly with decreasing intervals.

2) *Phase Compensation Factor Searching Algorithm*: To reduce the computational complexity of parameter searching, the Radon transform is operated in the Fourier domain [33]. By applying a DFT about α_1 on both sides of (16), we can obtain the following result:

$$H(\omega, \alpha_2, \beta_1) = \sum_{k=0}^{K-1} R(\omega, kDT_s) \exp\{j\omega\alpha_2 kDT_s\} \times \exp\left\{-j2\pi\left[\beta_1 kDT_s + \frac{1}{2}\alpha_2(kDT_s)^2\right]\right\} \quad (24)$$

where $H(\omega, \alpha_2, \beta_1)$ and $R(\omega, kDT_s) \exp\{j\omega\alpha_2 kDT_s\}$ are the DFT results about α_1 on $G(\alpha_1, \alpha_2, \beta_1)$ and $X(\alpha_1 + \alpha_2 kDT_s, kDT_s)$, respectively. ω can be expressed as the following with respect to i :

$$\omega_i = 2\pi MT_s(i-1)/N_{\alpha_1}, i = 0, 1, \dots, N_{\alpha_1} - 1. \quad (25)$$

For each searching variable m , (24) can be formed into a matrix expressed as

$$H = [H^1, H^2, \dots, H^{N_{\alpha_1}}]^T, H^i = L^i R^i \quad (26)$$

where the symbol “ T ” denotes a transpose. The dimension of H is $N_{\alpha_1} \times N_{\alpha_2}$. R^i is the data space with dimension of $K \times 1$. H^i is the model space with dimension of $N_{\alpha_2} \times 1$. L^i is the transform operator, whose size is $N_{\alpha_2} \times K$. The elements in R^i and L^i are given as

$$R_k^i = R(\omega_i, kDT_s) \quad (27)$$

$$L_{j,k}^i = \exp\{j\omega_i \alpha_2 j kDT_s\} \times \exp\left\{-j2\pi\left[\beta_{1m} kDT_s + \frac{1}{2}\alpha_{2j}(kDT_s)^2\right]\right\} \quad (28)$$

where $\exp\{-j2\pi\beta_{1m} kDT_s\}$ represents the phase compensation factor. By reducing the search interval $\Delta\beta_1$ defined in (23), the compensation of the first-order phase in (15) becomes as precise as possible. Thus, the detection performance is greatly improved.

The expected result can then be obtained by performing an inverse DFT on $H(\omega, \alpha_2, \beta_1)$

$$G(\alpha_1, \alpha_2, \beta_1) = \sum_{i=1}^{N_a} H(\omega, \alpha_2, \beta_1) \exp\{j\omega\alpha_1\}. \quad (29)$$

The 3-D matrix $G(\alpha_1, \alpha_2, \beta_1)$ can be converted into three 1-D vectors. Therein, what we are most concerned with is the 1-D line spectrum estimation result $G_S(\alpha_1)$ obtained by

$$G_S(\alpha_1) = \max_{\alpha_2, \beta_1} \left| \frac{1}{K} G(\alpha_1, \alpha_2, \beta_1) \right|^2. \quad (30)$$

To improve the detection performance of a single target to the best of our ability, $G_S(\alpha_1)$ is obtained according to the

Algorithm 1: Scheme of the LTCILSD.

- 1: **Procedure** LTCILSD.
 - 2: Partition the available input data record into K segments with overlap.
 - 3: Apply a time-domain window function to each segment.
 - 4: Apply a Fourier transform to each windowed segment.
 - 5: Take a DFT along the frequency bins for each segment.
 - 6: For each searching variable m
($m = 0, 1, \dots, N_{\beta_1} - 1$):
(a) Calculate the matrix H according to (25)–(28).
(b) Perform an inverse DFT on matrix H along the column.
 - 7: Calculate the 1-D discrete spectrum according to (30) in case of a single target or according to (31) in case of multiple targets.
 - 8: **End procedure**.
-

maximum energy point of $G(\alpha_1, \alpha_2, \beta_1)$. Compared with the multiresolution searching strategy, which takes the maximum of FFT across segments in the dimension α_1, α_2 where there is no signal, the method defined in (30) further increases the integration gain [23].

If multiple targets appear in the frequency range of interest, only the strongest target can be detected using (30). In this case, we can utilize the maximum-value projection method to detect multiple targets, which is defined as

$$G_M(\alpha_1) = \max_{\alpha_2, \beta_1} \left[\left(\left| \frac{1}{K} G(\alpha_1, \alpha_2, \beta_1) \right|^2 \right) (\alpha_2, \beta_1 | \alpha_1) \right]. \quad (31)$$

Each point in the $G_M(\alpha_1)$ represents its maximum value in the dimension α_2, β_1 in 3-D matrix $|(1/K)G(\alpha_1, \alpha_2, \beta_1)|^2$. Therefore, all information in the parameter domain is considered and the characteristics of each target can be retained.

3) *Summary of Major Steps*: Considering the above theoretical derivation, the implementation of the LTCILSD algorithm is summarized in Algorithm 1.

E. Analysis of the Computational Complexity

Generally, the Fourier transform may be efficiently realized by FFT. Define the complex multiplication and complex addition as I_m and I_a , respectively. The computational complexity of the periodogram, CAPSE, PRPFT (see Appendix II for its implementation), and the proposed algorithm is shown in Table I. The complexity of the periodogram and CAPSE method is the same, and the proposed method is slightly higher than that of the PRPFT method. Apparently, the complexity of the proposed method is dependent mostly on the number of search parameters. We can reduce the computational amount by limiting the search range of the parameters according to the DFT results of signal segments. In addition, the search intervals of parameters α_2 and β_1 can be reasonably selected to decrease the computational complexity according to the coherent accumulation results.

TABLE I
COMPLEXITY OF PERIODOGRAM, CAPSE, PRPFT, AND LTCILSD

Methods	Complexity
Periodogram	$\left[\frac{N}{2} \log_2 N \right] I_m + [N \log_2 N] I_a$
CAPSE	$\left[\frac{N}{2} \log_2 N \right] I_m + [N \log_2 N] I_a$
PRPFT	$\left[\frac{N}{2} \log_2 M + N_{\alpha_1} N_{\alpha_2} K \left(\frac{\log_2 K}{2} + 1 \right) \right] I_m + [N \log_2 M + N_{\alpha_1} N_{\alpha_2} K \log_2 K] I_a$
LTCILSD	$\left[N \log_2 M + N_{\beta_1} N_{\alpha_2} N + \frac{N_{\alpha_1} N_{\alpha_2}}{2} \log_2 N_{\alpha_1} \right] I_m + [2N \log_2 M + N_{\alpha_1} N_{\alpha_2} \log_2 N_{\alpha_1}] I_a$

F. Analysis of Coherent Integration Gain

By substituting (7) and (13)–(15) into (16), when the searching parameters α_2 and β_1 are exactly matched to the signal chirp rate $2\mu_2$ and equivalent initial frequency μ_1^* , the spectrum estimated by the proposed algorithm can be expressed as

$$G_x(a) = \{A \text{sinc}[\pi M T_s[(\mu_1 - \alpha_1)]]\}^2. \quad (32)$$

The Doppler effect causes a broadening of the spectral beam of the received signal. In other words, the energy of the signal is dispersed in a certain frequency band rather than concentrated on one frequency bin. From (32), it is evident that the energy can be focused perfectly on the initial frequency of the signal by applying the LTCILSD method. In this sense, the proposed method can be regarded as an expansion of the periodogram method for the case of a moving target.

Given that the received data of passive sonar is composed of a signal [expressed as (3)] and additive white noise with zero mean, the mean of the spectrum results for additive noise estimated by the proposed method can be readily derived (see Appendix III) and is expressed as

$$E[G_n(l)] = \frac{P_n}{K} + \frac{P_n}{K^2} \sum_{\substack{k=0, q=0 \\ k \neq q}}^{K-1} \rho(k, q) \quad (33)$$

where $\rho(k, q)$ denotes the correlation coefficient between the k^{th} and q^{th} noise segments and is expressed as

$$\begin{aligned} & \rho(k, q) \\ &= \frac{\sin[\pi(l_q - l_k)(M - |k - q|D)/M]}{M \sin[\pi(l_q - l_k)/M]} \\ & \times e^{j\pi(M - |k - q|D - 1)(l_q - l_k)/M} e^{j2\pi(k - q)Dl_q/M} e^{j(\Psi_{q, l_q} - \Psi_{k, l_k})} \\ & k = 0, 1, \dots, K - 1; q = 0, 1, \dots, K - 1; k \neq q. \end{aligned} \quad (34)$$

Here, $M - |k - q|D > 0$ must be satisfied and Ψ_{k, l_k} represents the compensated phase of k^{th} segment.

If we define the gain of the signal processor as the difference between output and input SNR, then the gain of the LTCILSD

method is given by

$$\begin{aligned} G_{\text{LTCIPSE}} &= 10 \log_{10}(KT) \\ &- 10 \log_{10} \left[\text{Re} \left(K + \sum_{\substack{k=0, q=0 \\ k \neq q}}^{K-1} \rho(k, q) \right) \right]. \end{aligned} \quad (35)$$

Here, we assume that the noise segments have identical Gaussian distributions and are statistically independent when there is no overlap between adjacent noise segments. In this case, $\rho(k, q) = 0$ for all u . However, the calculation of $\rho(k, q)$ is complicated when time slices are overlapped. Therefore, considering two situations for segmenting the data (partial overlap and no overlap), the gains for the LTCILSD approach are analyzed as follows.

1) *Segment the Data Without Overlap*: In this case, the expression of (35) can be simplified as

$$G_{\text{LTCIPSE}} = 10 \log_{10} T. \quad (36)$$

A comparison of processing gains for the LTCILSD, periodogram, CAPSE [23], and PRPFT [28] methods is illustrated in Fig. 2(a). The act of taking the maximum in the FFT makes an equivalent gain between PRPFT and CAPSE. When the time duration is 1 s, the four methods are equivalent where there is no gain. When the time duration increases, both the periodogram and LTCILSD methods have the highest gain. The gain of the LTCILSD method is approximately 10 dB higher than that of the CAPSE and PRPFT methods when the time duration is 1000 s.

2) *Segment the Data With Partial Overlap*: Let us take an example and observe the changes in the gain with increasing overlap. By considering $T = 1000\text{s}$ and $MT_s = 1\text{s}$, each one partially overlapping in conformity with the fractional overlap r , the gain with respect to r is shown in Fig. 2(b). By adopting the overlap of adjacent time slices, a further increase in the effective number of segments is achieved, which is also tightly linked to the more correlated time slices. Thus, the gain exhibits only

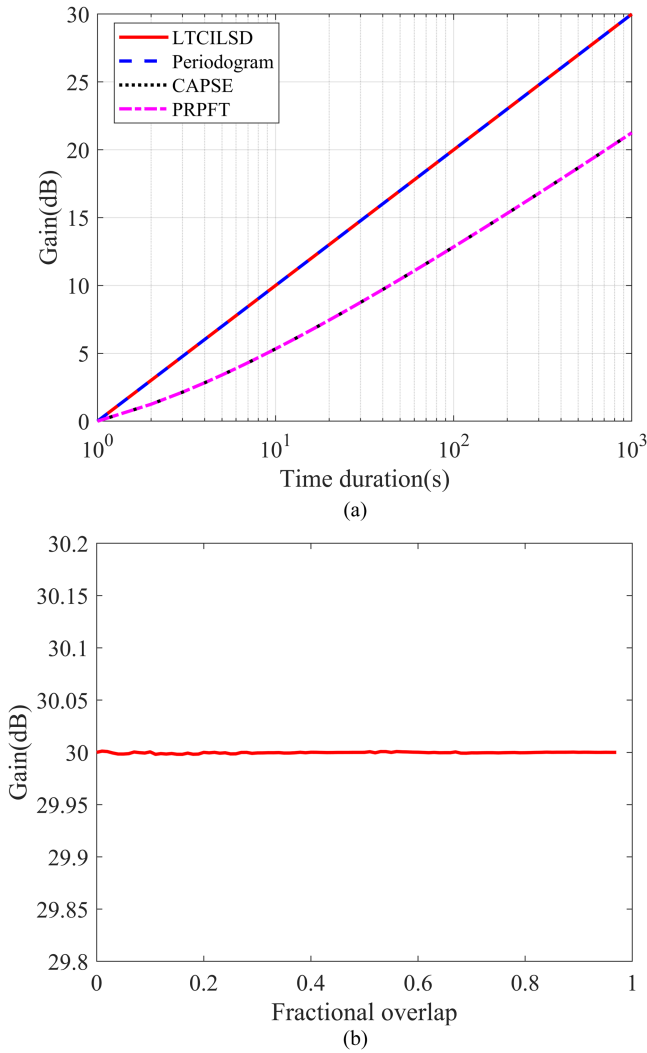


Fig. 2. (a) Comparison of gain obtained by LTCILSD, periodogram, CAPSE, and PRPFT methods. (b) Gain marked in terms of fractional overlap r for the LTCILSD method with a fixed record length and segment duration.

minor fluctuations around 30. This implies that no gain or loss occurs at any overlap ratio.

From this analysis, it can be concluded that, compared with the method of dividing the record without overlap, the LTCILSD method with overlap cannot achieve additional processing gains. Therefore, to reduce the computational effort, no overlap is adopted for the subsequent simulations and trial data processing.

Acoustic signals fluctuate with time when they propagate through a dynamic medium such as the ocean. This fluctuation can be characterized in terms of the temporal coherence. The temporal coherence of acoustic signals determines the signal gain for phase-sensitive signal processing algorithms, such as the LTCILSD method proposed in this article. Various inhomogeneities and acoustic interference phenomena in the ocean distort the acoustic signal randomly, which limits the signal coherence time. In particular, the signal received by a stationary receiver is compressed or stretched if the source is moving. This phenomenon further decreases the signal coherence time.

Therefore, in Section IV-D, the effect of the underwater acoustic channel on the processing gain of the coherent integration algorithm is analyzed in detail by processing the experimental data.

III. SIMULATION PERFORMANCE

In this section, the impact of the Doppler shift on tone detection and the effectiveness of the proposed method are verified by simulating.

A. Performance Comparison Under Different Doppler Effects

To verify the proposed LTCILSD method, three simulations under different Doppler effects were considered. The simulated parameters are listed in Table II (NA indicates not available). The sampling rate was 10 kHz, and the total length of the data to be analyzed was 100 s. The segment length was selected as $T_0 = 20$ s according to (6). Fig. 3 illustrates the line spectrum estimation results for different simulations.

For the stationary target case in Simulation 1, the periodogram achieves the best performance with no scalloping loss because its frequency resolution is the highest among all methods. Due to the discretization of FFT analysis, there is a phase difference between adjacent segments. In Fig. 3(a), the spectrum estimated by the CAPSE method conforms with that of the PRPFT method because the principles used to compensate the phase difference are identical. The proposed method can obtain a narrower beamwidth because the more accurate phase difference compensation disperses the signal energy less at other frequencies.

Considering the moving target cases in Simulations 2 and 3, the energy of the target is defocused to approximately 0.12 and 0.22 Hz, respectively. A comparison of the line spectrum estimation results in Fig. 3(b) and (c) reveals that the proposed method can provide the highest frequency aggregation and the lowest noise level under different Doppler effects. This implies that the detection performance of the proposed algorithm is independent on the target velocity. Therefore, this method can be applied to detect targets with arbitrary velocities.

B. Performance Analysis Under Multitarget Scenario

The simulated parameters are shown in Table III. It can be seen from Fig. 4(a) that the Doppler effect of Target 1 is the most severe (changing from 350.4 to 349.7 Hz), followed by that of Target 2, and the frequency of Target 1 fluctuates slightly. The frequency lines of Target 1 and 3 intersect each other. To make matters worse, they are too close to one another to distinguish their initial frequencies. Compared with Target 1 and Target 3, Target 2 exhibits a much lower energy and is difficult to discern in the low-frequency analysis and recording (LOFAR) result.

The detection results shown in Fig. 4(b) indicate that the proposed method can provide a lower sidelobe than the other methods, thereby allowing it to distinguish Target 1 from Target 3, which is difficult for the other methods. In addition, Target 2 cannot be detected by periodogram, CAPSE, and PRPFT because it exhibits a relatively serious Doppler shift, and its

TABLE II
PARAMETER DESIGNED FOR DIFFERENT SIMULATIONS

Parameter	Simulation 1	Simulation 2	Simulation 3
r_0 (m)	NA	780	780
θ ($^\circ$)	NA	70	70
v (m/s)	NA	2.5	5
f_0 (Hz)	100.02	100	100
Input SNR (dB)	20	20	20

TABLE III
SIMULATED PARAMETERS OF DIFFERENT TARGETS

Parameter	Target 1	Target 2	Target 3
r_0 (m)	780	1500	3000
θ ($^\circ$)	70	65	20
v (m/s)	5	4	1
f_0 (Hz)	350	353	350
Input SNR (dB)	10	2	10

energy is extremely low. However, the proposed method can focus the energy of Target 2 perfectly.

C. Detection Performance Evaluation

To evaluate the performance of the proposed LTCILSD detector, we used the constant false alarm rate (CFAR) detection algorithm described in [28] to compare the performance of the four investigated methods. The probability of false alarms was set to $P_{FA} = 0.01$. The given integer was $M = 50$. We performed the Monte Carlo simulation 100 times to obtain the detection results.

In this section, the simulated parameters were consistent with Simulations 1 and 3 listed in Table II. Except that the source frequencies were both set to 100 Hz. The probability of detection P_D versus SNR curves of the different algorithms are shown in Fig. 5.

From Fig. 5(a), where the source and receiver are stationary, it is shown that the $P_D - SNR$ curves for the different methods are quite close to each other. When the Doppler shift occurs, the results shown in Fig. 5(b) suggest that the achievable P_D values of the periodogram and CAPSE methods have rather large gaps compared with that of the LTCILSD detector. For a given P_D value (e.g., $P_D = 0.8$), the input SNR demanded by the PRPFT method is approximately 3 dB higher than that of the LTCILSD processor. By comparing Fig. 5(a) with (b), it can be observed that the gain losses of periodogram, CAPSE, PRPFT, and LTCILSD methods caused by Doppler shift are 8, 9, 4, and 2 dB, respectively. The LTCILSD detector therefore has

the smallest gain loss and shows good robustness for moving target detection.

Given these comparative simulation results, the superiority of LTCILSD in detecting moving targets was demonstrated. The SNR threshold of this method is the lowest among the four evaluated methods, and its detection probability can reach a value of 1 even at a negative SNR.

IV. REAL DATA ANALYSIS

To verify the robustness of the proposed LTCILSD algorithm for detecting targets under the influence of the Doppler effect, we processed the data collected from two sea trials. The detection results were compared with those of the other evaluated methods. Furthermore, the influence of the underwater acoustic channel on the maximum coherent integration time was also analyzed.

A. Description of the Experiments

The first experiment (referred to as E1) was conducted in the South China Sea on July 12, 2021, as shown in Fig. 6. The vertical line array (VLA) was composed of 32 elements with an interval of 2 m. The top element, labeled #1, was deployed at a depth of 275 m, and the bottom element, labeled #32, was deployed at approximately 337 m. A transmitting transducer was carried by a test vessel to transmit a pure tone. The depth of the transducer was 20 m. The distance between the VLA and transmitter was 5.02 km.

Data from the second experiment (denoted as E2) was obtained using a horizontal line array (HLA North) in SWellEx-96

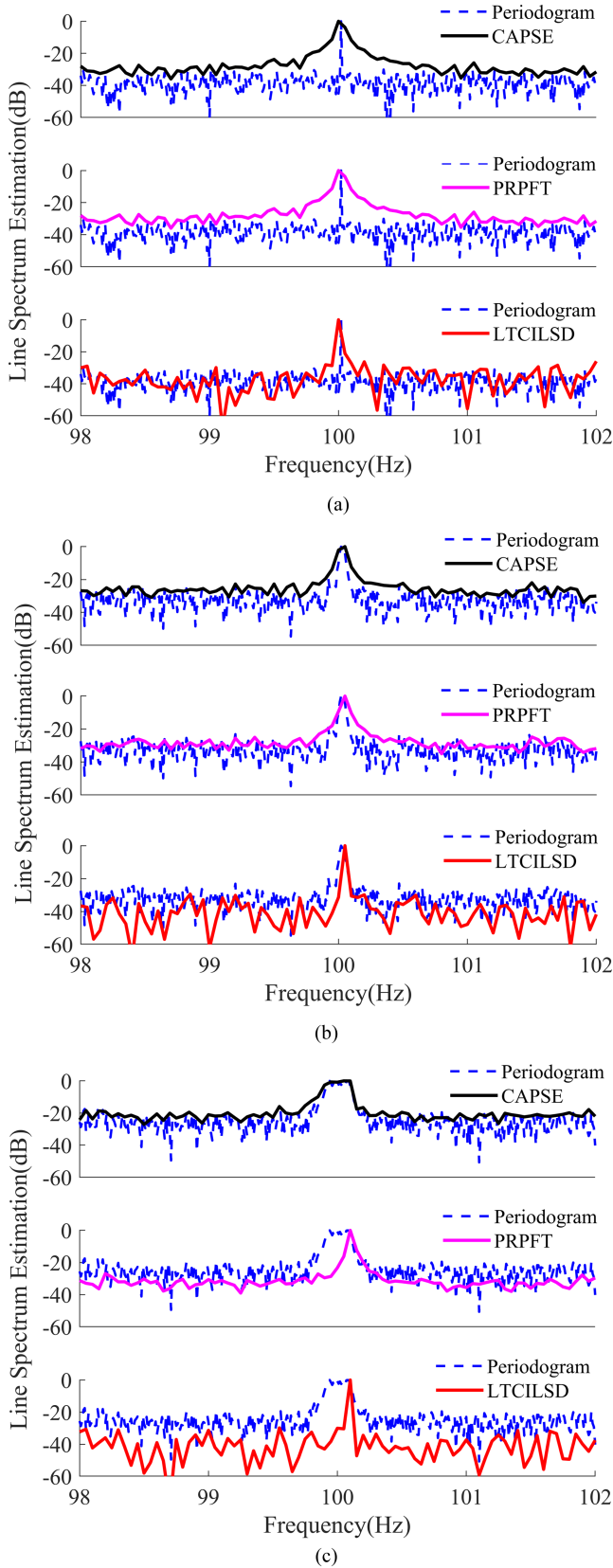


Fig. 3. Detection results of the periodogram, CAPSE, PRPFT, and LTCILSD methods under different Doppler effects shown in (a) simulation 1, (b) simulation 2, and (c) simulation 3.

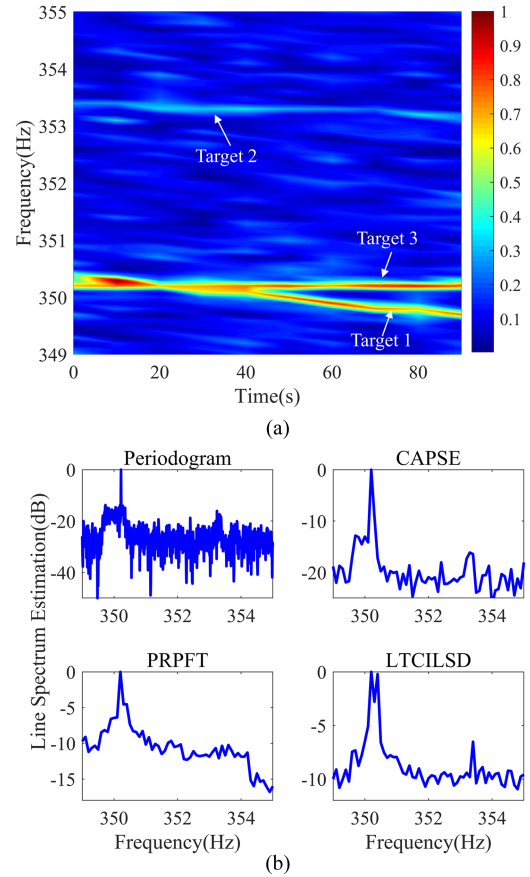


Fig. 4. Simulated results under multitarget scenario. (a) LOFAR of the simulated signals. (b) Comparison of the periodogram, CAPSE, PRPFT, and LTCILSD methods.

Event S5. The source ship (R/V Sproul) started its track south of HLA North and proceeded northward at a speed of 2.5 m/s. Acoustic sources transmitted numerous tones of various source levels between 49 and 400 Hz.

B. Detection Results for Different Doppler Frequency Shifts

The data processed in this section is listed in Table IV. The value of the Doppler frequency shift is directly proportional to the transmission frequency of the source at a constant radial velocity of the target. In other words, higher-frequency tones yield larger Doppler frequency shifts. Therefore, the Doppler shift of Data sets 1–3 increases accordingly. The search scope of α_2 is set as $-0.005 \leq \alpha_2 \leq 0.005$. We define the search steps as $\Delta\alpha_2 = 0.0001$ and $\Delta\beta_1 = 0.001$. Figs. 7–9 illustrate the detection results for a single element with different data.

The red dashed lines in Figs. 7(b), 8(b), and 9(b) represent the noise levels of the different methods. Each line was calculated by averaging all the power in the frequency band of interest, excluding the frequency corresponding to the maximum of the estimated spectrum.

As shown in Fig. 7(a), the signal is too weak to be clearly observed in the LOFAR result. Only a few discontinuous

TABLE IV
DATA SETS OF DIFFERENT DOPPLER FREQUENCY SHIFTS

	Data set 1	Data set 2	Data set 3
Experiment	E1	E2	E2
Velocity of the source (m/s)	0	2.5	2.5
Source frequency (Hz)	27	67	341
Source level (dB)	NA	132	132
Recorded time length (s)	600	600	600
Sampling rate (Hz)	10000	3276.8	3276.8

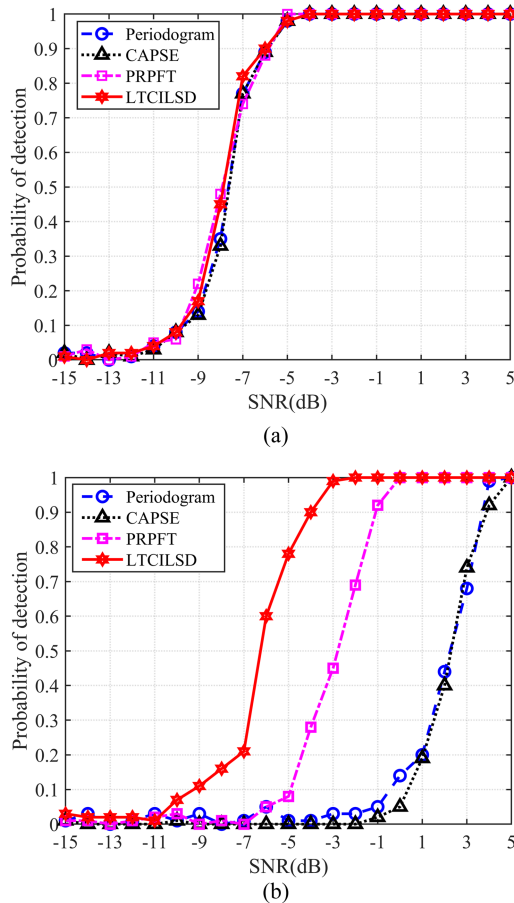


Fig. 5. Relationship between the probability of detection and SNR of different algorithms when the velocity of the target is (a) $v = 0$ and (b) $v = 5$ m/s.

frequency lines around 27 Hz exist from 0 to 600 s. In addition, the signal exhibits a minor frequency fluctuation due to the acoustic propagation in complex underwater acoustic channel. This frequency fluctuation may significantly decrease the gain of the coherent integration methods. From the detection results shown in Fig. 7(b), it is difficult to identify the target manually in the periodogram because a cluster of peaks appears around 27 Hz. This deterioration of the detection performance could be attributed to the high frequency resolution of the periodogram and the frequency shift due to the dynamic ocean fluctuation.

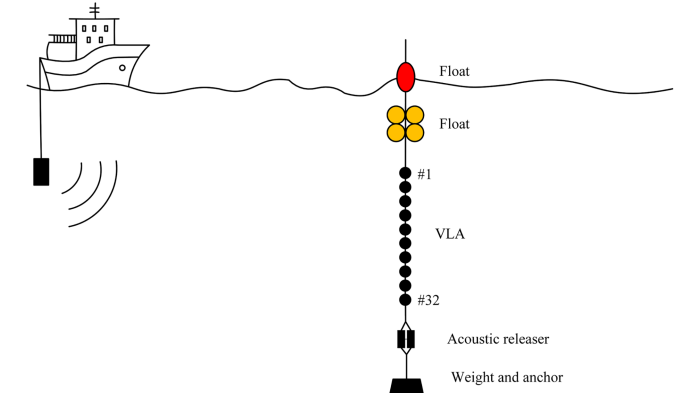


Fig. 6. Schematic of the experimental situation.

Therefore, a coherent integration gain is not satisfactorily obtained by the periodogram method. Conversely, the lower frequency resolution of the CAPSE method generates a lower sensitivity to the frequency shift. Therefore, the peak around 27 Hz can help identify the signal correctly. It is evident that the PRPFT and LTCILSD methods both provide better spectrum estimation results. According to the noise levels shown in Fig. 7(b), the extra gain obtained by the LTCILSD method is 9.3 dB and 15.3 dB compared to the PRPFT and CAPSE methods, respectively.

In Fig. 8, the frequency of the received signal changes slowly. Only intermittent frequency lines are visible because of the strong background noise. This problem cannot be solved efficiently through the periodogram method, because the tone is too weak to be extracted. LTCILSD, in comparison to the PRPFT and CAPSE methods, performs relatively well in frequency resolution improvement, and the detection performance increases by 5.6 dB and 18.6 dB, respectively.

The LOFAR shown in Fig. 9(a) indicates that the instantaneous frequency of the received data changes from 341.4 to 340.6 Hz during the recorded time length of 600 s. Tone detection is affected by strong interference arising from 350 to 450 s in a range of 1 Hz, i.e., from 341 to 342 Hz. The upper graphs in Fig. 9(b) show that the tone cannot be detected effectively through the periodogram and CAPSE approaches, because the energy of the tone is dispersed over a wide frequency band. In contrast, the tone is considerably enhanced by the PRPFT and

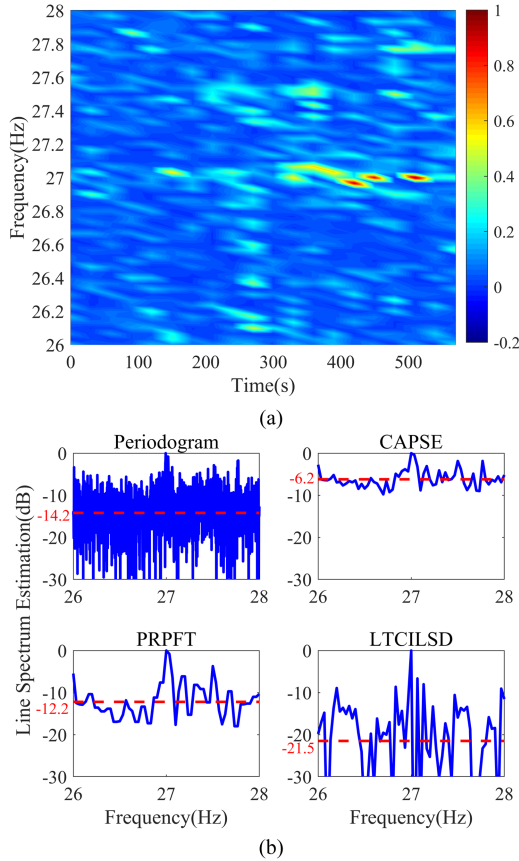


Fig. 7. Detection results of data set 1 depicted by (a) LOFAR of the original received data and (b) comparison of spectra obtained by periodogram, CAPSE, PRPFT, and LTCILSD methods.

LTCILSD methods, as shown in the lower graphs in Fig. 9(b). Compared with PRPFT method, an additional gain of 5.8 dB in detection performance is obtained by the proposed method.

By comprehensively comparing the processing results for different Doppler frequency shifts, we found that the proposed algorithm can focus the vast majority of the signal energy on the initial frequency. Furthermore, it provides a higher SNR gain and a stronger background noise weakening effect than other methods even under a severe Doppler frequency shift.

C. Detection Results Under Multitarget Scenario

The detection performance of the proposed method for multiple moving targets was validated using the data collected in E2. We refer to the “shallow” and the “deep” sources as Targets 1 and 2, respectively. The processing results in the frequency band of 333–342 Hz are shown in Fig. 10.

In Fig. 10(a), there are three tones, i.e., the 335-Hz tone emitted by Target 1, the 338-Hz Tone 1 with 158-dB source level emitted by Target 2 and the 341-Hz Tone 2 with 132-dB source level emitted by Target 2. Due to the Doppler effect, these three tones all have a large Doppler frequency shift. In particular, Tone 2 of Target 2 cannot be easily distinguished in the LOFAR results due to its extremely low energy.

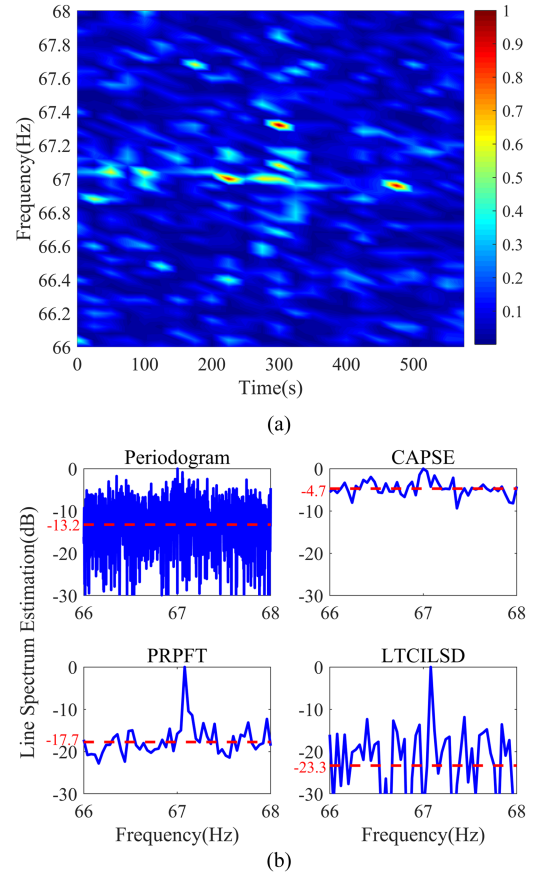


Fig. 8. Detection results of data set 2 depicted by (a) LOFAR of the original received data and (b) comparison of the periodogram, CAPSE, PRPFT, and LTCILSD methods.

In Fig. 10(b), four methods are able to detect Target 1 and Tone1 of Target 2. However, the spectra estimated by the periodogram and CAPSE methods exhibit obvious frequency expansion. Although the PRPFT method can coherently gather the energy of the strong tones, it cannot detect the low-energy Tone 2 of Target 2. The proposed method can compensate the phase difference more accurately using the phase compensation factor searching algorithm, such that the weak tones can be detected robustly under the influence of strong tones.

We may conclude that the proposed phase compensation factor searching algorithm can achieve more accurate phase compensation than PRPFT, which gives it the potential to detect weak tones robustly in the presence of strong tones.

D. Estimation of Coherent Integration Time

As mentioned previously, the Doppler shift caused by the target motion induces coherence degradation. This effect was analyzed in detail by comparing the coherent integration gain obtained under different severities of the Doppler effect. The processing results of data recorded from sea trials with different source–receiver geometries were compared.

Fig. 11 shows a simplified diagram of four types of source–receiver geometry, including trial Tr1 with a fixed source, trial Tr2 with a radial mobile source, trial Tr3 with a transverse

TABLE V
 PARAMETERS OF SEA TRIALS

Source-receiver geometry	Source frequency (Hz)	Recorded time length (s)	Source- receiver distance (km)	Source velocity (m/s)
Tr1	71	600	9.0	0
Tr2	150	600	8.2–6.0	2.5
Tr3	71	600	5.5–4.0	2.5
Tr4	335	600	1.0–0.7–1.0	2.5

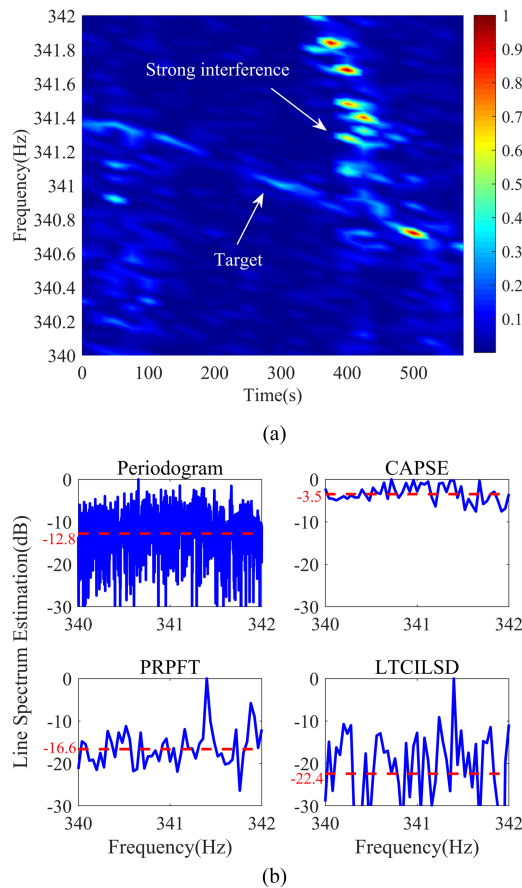


Fig. 9. Detection results of data set 3 depicted by (a) LOFAR of the original received data and (b) comparison of the periodogram, CAPSE, PRPFT, and LTCILSD methods.

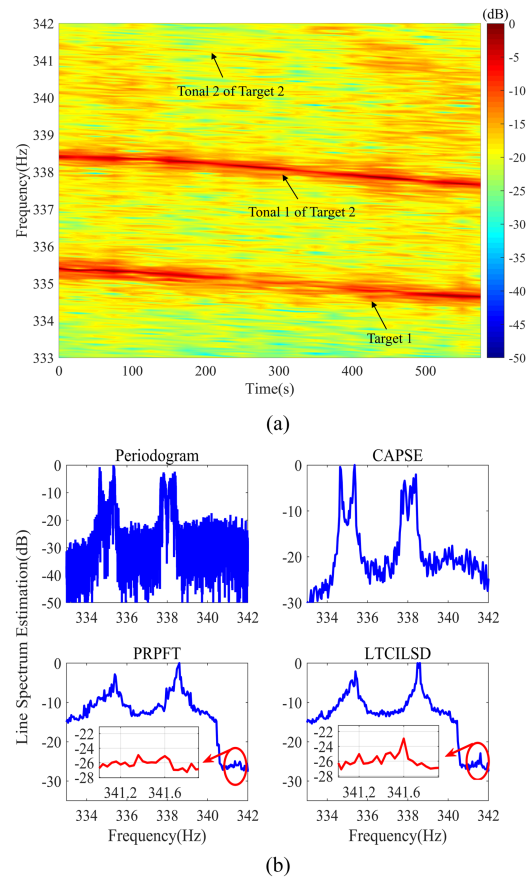


Fig. 10. Processing results for two targets in E2 depicted by (a) LOFAR of the original received data and (b) comparison of the periodogram, CAPSE, PRPFT, and LTCILSD methods.

mobile source, and trial Tr4 with a transverse mobile source passing over the receiver. The trials Tr1 to Tr3 are from E1, and Tr4 is from E2. The motion tracks of the sound source during the observation time are described by the red solid lines, the distances between the source and the receiver at the beginning and end of the track are described by the black dashed lines, and the closest distance between source and receiver for Tr4 is also marked in Fig. 11(d). The recorded parameters of the four sea trials are shown in Table V.

For each element in the receiving array for each trial, the four methods discussed previously were used to observe the processing gain versus integration time. The segment length was

set to 10 s. The time duration was set to 30 to 600 s with an interval of 10 s. The detailed procedure for estimating the output SNR was as follows.

- Determine the time duration.
- Estimate the spectrum of the collected data using the coherent integration methods.
- Estimate the power of the tone by determining the maximum of the signal spectrum.
- Intercept the noise data collected at sea with the same time duration from the data start time.
- Estimate the noise power spectrum of the noisy data through the coherent integration methods.

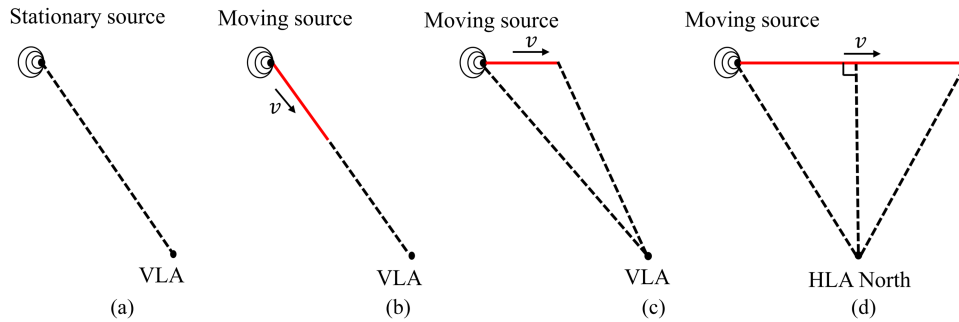


Fig. 11. Source-receiver geometries of (a) trial Tr1 with a fixed source, (b) trial Tr2 with a radial moving source, (c) trial Tr3 with a transverse moving source, and (d) trial Tr4 with a transverse moving source passing over the receiver.

TABLE VI
ROUGH ESTIMATION OF THE SIGNAL COHERENCE TIME USING DIFFERENT METHODS

Source-receiver geometry	Periodogram (s)	CAPSE (s)	PRPFT (s)	LTCILSD (s)
Tr1	> 600	> 600	> 600	> 600
Tr2	250	250	250	440
Tr3	220	220	220	290
Tr4	30	30	30	190

- vi) Estimate the power of the noise component by obtaining the mean of the noise power spectrum.
- vii) Compute the output SNR by dividing the estimated power of the tone by the estimated power of the noise component.

Given that no noise data were collected in Tr4, the power of the noise component was estimated by averaging all the power in the interesting band, excluding the frequency corresponding to the maximum of the signal spectrum obtained in step (ii). When the time duration is determined to be 1s, the output SNR estimated by the periodogram can be regarded as the original SNR of the collected data, i.e., the input SNR of all coherent integration methods. We defined the gain of the signal processor as the difference between the output and input SNR. Average the gains of all elements of the receiving array in each trial. The gain comparison of the theoretical curve and periodogram, CAPSE, PRPFT, LTCILSD methods is shown in Fig. 12. It can be seen that the difference between the gain of the coherent integration algorithms and the theoretical value increases as the Doppler effect becomes more severe.

The signal coherence time determines the maximum time the signal can be integrated before losing coherence, which can be defined as the time duration corresponding to the maximum temporal processing gain. From the gain curves plotted in Fig. 12, the signal coherence time durations estimated by different methods are listed in Table VI.

If the target is stationary, the energy of incoherent scattering is commonly neglected. The ocean acoustic channel can be modeled as a slowly time-varying coherent multipath channel. Previously published studies have demonstrated that the coherence length is at least a few minutes [35], [36], [37].

From Fig. 12(a), as one would expect, the gains of all methods increase as the time duration increases. A comparison of the four curves shows that the CAPSE method provides the same gain level as the PRPFT method, and the gain of the LTCILSD method is slightly lower than that of the periodogram. Therefore, this result implies that the signal coherence time is at least 600 s when the source is fixed.

If the target is moving, the slowly time-varying coherent multipath channel model is invalid. The coherence integration gain decreases because of the Doppler effect. As shown in Fig. 12(a)–(c), the maximum gain for each method is approximately 2–3 dB lower compared to that of the previous subfigure. Even worse, compared with the results in Tr1, the gain level of the periodogram decreases by at least 13 dB in Tr4. It can be seen from Table VI that the signal coherence time of each approach decreases quickly compared to that of a fixed source. The results indicate that the coherence degradation induced by target motion is intolerable, especially when there is a relatively large Doppler shift (i.e., in Tr4). It is notable that the proposed method can achieve robust performance under different severities of the Doppler effect. Furthermore, for Tr2, Tr3, and Tr4, the signal coherence time estimated by the proposed method is the longest among all the methods.

From the experimental data processing results, it may be concluded that the coherence degradation depends on the severity of the Doppler effect. Compared to other methods, the detection performance degradation of the LTCILSD algorithm is the smallest. Therefore, the signal over a longer time duration can be integrated coherently by the proposed method, resulting in higher coherent integration gain.

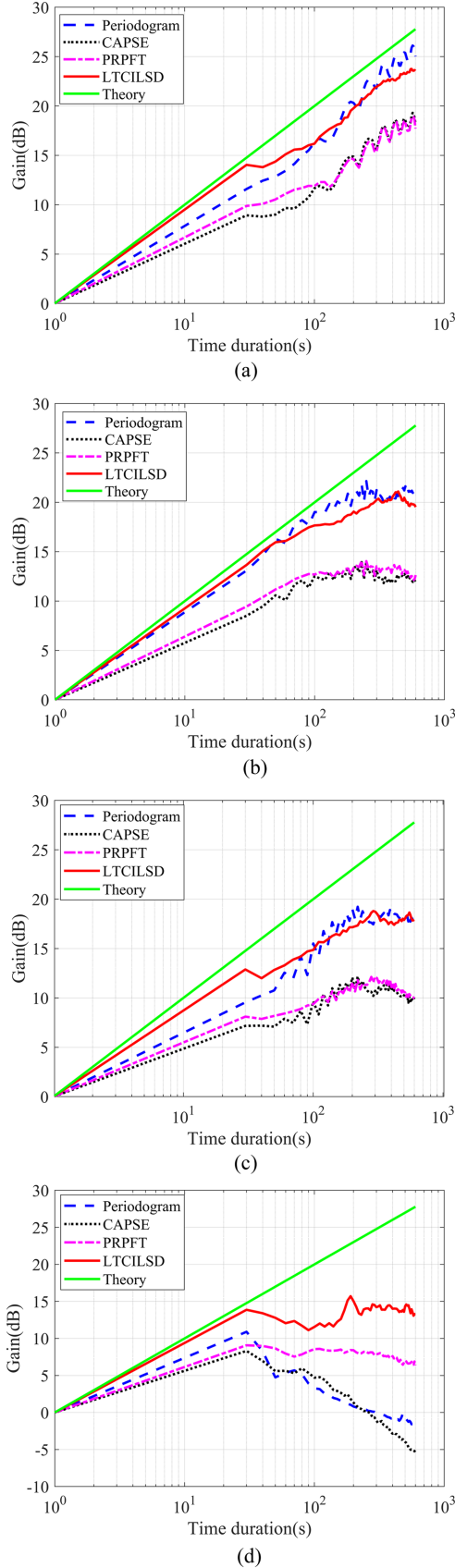


Fig. 12. Coherent integration gain obtained by processing the collected data, plotted as a function of time duration for (a) Tr1, (b) Tr2, (c) Tr3, and (d) Tr4.

V. CONCLUSION

In this article, a narrowband line spectrum detection method was proposed to realize long-time coherent integration under the influence of Doppler shift. We introduced the condition of modeling received signal as a polynomial-phase signal in the frequency domain, i.e., the frequency variation of each data segment should be smaller than a frequency resolution cell. Using the proposed LTCILSD method, the time-variant frequency drift and phase difference between segments were compensated simultaneously. Thus, we were capable of achieving a long-time coherent integration in the time-frequency domain to enhance the tonal signal radiated from moving targets. The proposed method was shown to perform better than the periodogram and CAPSE methods. It could detect the moving target robustly even under the influence of serious Doppler effect. Compared with the PRPFT method, the proposed algorithm improved the precision of the phase compensation using the phase compensation factor searching algorithm. Thus, higher coherent integration gain and frequency aggregation could be obtained by the proposed method. This enabled it to obtain better detection performance in a multitarget scenario.

The temporal and spatial variations of the underwater acoustic channel and the relative motion between the source and receiver distort acoustic signals, thereby reducing the signal coherence time. Experimental results show that the proposed algorithm can compensate this distortion and extend the signal coherence time to hundreds of seconds. Therefore, it is suitable for detecting the tones with Doppler frequency shifts in complex underwater situations.

APPENDIX I

Taking the derivative of the instantaneous frequency of the received signal in (2) with respect to time t , we have

$$f'_I(t) = \frac{df_I(t)}{dt} = -\frac{f_0 v^2}{1 - \frac{v^2}{c^2}} \frac{r_0^2 \sin^2(\theta)}{c \left[v^2 \left(t - \frac{r_0 \cos(\theta)}{v} \right)^2 + r_0^2 \sin^2(\theta) \right]^{\frac{3}{2}}}. \quad (37)$$

Given that the speed of an underwater moving target is considerably less than the speed of sound in water, i.e., $v^2 \ll c^2$. So $1 - v^2/c^2 \approx 1$. Furthermore, if we assume

$$v^2 \left(t - \frac{r_0 \cos(\theta)}{v} \right)^2 \ll r_0^2 \sin^2(\theta)$$

(37) can be approximated as a constant. This means that the instantaneous frequency varies linearly.

Define the scale factor ε as

$$v^2 \left(t - \frac{r_0 \cos(\theta)}{v} \right)^2 = \varepsilon r_0^2 \sin^2(\theta). \quad (38)$$

In order to explore the effect of the linear approximation on the coherent integration gain, the gains are calculated using a different scale factor ε . The simulation parameters are listed in

TABLE VII
SIMULATION PARAMETERS OF RECEIVED SIGNAL

Parameters	r_0 (km)	θ (°)	v (m/s)	T (s)	f_s (Hz)	Input SNR (dB)
Value	3	10	2	3000	1000	20

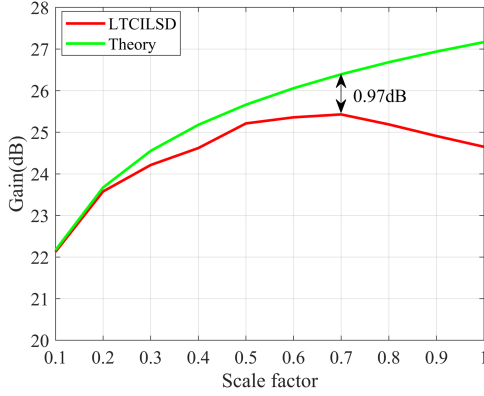


Fig. 13. Gain comparison of the theory and LTCILSD method with different scale factors.

Table VII. The detailed procedure for calculating the gain refers to Section IV-D. Except that the analyzed signal corresponding to each ε is intercepted as that located in the time interval

$$t \in \left[\frac{r_0 \cos(\theta)}{v} - \frac{\sqrt{\varepsilon} r_0 \sin(\theta)}{v}, \frac{r_0 \cos(\theta)}{v} + \frac{\sqrt{\varepsilon} r_0 \sin(\theta)}{v} \right].$$

Fig. 13 shows the gain comparison of the theory and the LTCILSD method with different scale factors. It can be found that when $\varepsilon \leq 0.7$, the gain of the LTCILSD increases with the increase of ε . Furthermore, the maximum gain loss compared with the theoretical gain is no more than 1 dB. Instead, when $\varepsilon > 0.7$, the gain of the LTCILSD decreases with the increase of ε . In the same way, the difference between theoretical gain and LTCILSD is increasing. This implies that the linear approximation is no longer applicable in the corresponding time intervals. Therefore, it can be concluded that as long as $\varepsilon \leq 0.7$, the linear approximation of the instantaneous frequency is reasonable, and the gain loss is negligible.

APPENDIX II

According the signal model established in this study, the implementation of the PRPFT [28] is summarized in Algorithm 2.

Algorithm 2: Scheme of the PRPFT.

- 1: **Procedure** PRPFT.
- 2: Partition the available input data record into K segments with overlap.
- 3: Apply a time-domain window function to each segment.

- 4: Apply a Fourier transform to each windowed segment.
- 5: For each $i, i = 0, 1, \dots, N_{\alpha_1} - 1$
 - Set $\alpha_{1i} = \alpha_{1\min} + i/MT_s$ according to (21)
 - For each $j, j = 0, 1, \dots, N_{\alpha_2} - 1$
 - Set $\alpha_{2j} = \alpha_{2\min} + j\Delta\alpha_2$ according to (22)
 - For each $k, k = 0, 1, \dots, K - 1$
 - Calculate $\rho(k) = X(\alpha_{1i} + \alpha_{2j}kDT_s, k) \exp\{-j\pi\alpha_{2j}(kDT_s)^2\}$
 - End
 - Calculate $G(\alpha_{1i}, \alpha_{2j}) = \max(|FFT(\rho)|)$ where $FFT(\cdot)$ denotes the Fast Fourier transform applied to the vector in $(\cdot)$.
 - End
- End
- 6: Calculate the one-dimensional discrete spectrum according to $G_S(\alpha_1) = \max_{\alpha_2} |G(\alpha_1, \alpha_2)/K|^2$ in case of a single target or according to $G_M(\alpha_1) = \max_{\alpha_2} [(|G(\alpha_1, \alpha_2)/K|^2)(\alpha_2|\alpha_1)]$ in case of multiple targets.
- 7: **End procedure.**

APPENDIX III

The mean of the line spectrum results for additive noise estimated by the proposed method is derived as follows.

The additive complex Gaussian white noise in k^{th} segment is expressed as $n_{k,m}, k = 0, 1, \dots, K - 1, m = 0, 1, \dots, M - 1$, where m represents the sample index. The DFT result of k^{th} noise segment is $N_{k,l}, k = 0, 1, \dots, K - 1, l = 0, 1, \dots, M - 1$, where l represents discrete frequency index. The long-time coherent integration is performed on discrete frequency bin l_k for the k^{th} noise segment. Assuming that $E[N_{k,l_k} N_{k,l_k}^*] = P_n$, the mean of the LTCILSD estimation for noise is given as (39) shown at the top of the next page, where $M - |k - q|D > 0$ must be satisfied and Ψ_{k,l_k} represents the compensated phase of the k^{th} segment.

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$$\begin{aligned}
E[G_n(l)] &= E \left[\left(\frac{1}{K} \sum_{k=0}^{K-1} N_{k,l_k} e^{-j\Psi_{k,l_k}} \right) \left(\frac{1}{K} \sum_{k=0}^{K-1} N_{k,l_k} e^{-j\Psi_{k,l_k}} \right)^* \right] \\
&= \frac{1}{K^2} E \left[\sum_{k=0}^{K-1} N_{k,l_k} N_{k,l_k}^* + \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} N_{k,l_k} N_{q,l_q}^* e^{-j\Psi_{k,l_k}} e^{j\Psi_{q,l_q}} \right] \\
&= \frac{1}{K^2} \left\{ KP_n + \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} E \left[N_{k,l_k} N_{q,l_q}^* \right] e^{-j\Psi_{k,l_k}} e^{j\Psi_{q,l_q}} \right\} \\
&= \frac{1}{K^2} \left\{ KP_n + \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} E \left[\frac{1}{M} \sum_{m=0}^{M-1} n_{k,m} e^{-j2\pi m l_k / M} \frac{1}{M} \sum_{m'=0}^{M-1} n_{q,m'}^* e^{j2\pi m' l_q / M} \right] e^{-j\Psi_{k,l_k}} e^{j\Psi_{q,l_q}} \right\} \\
&= \frac{1}{K^2} \left\{ KP_n + \frac{1}{M^2} \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} E \left[\sum_{m=0}^{M-|k-q|D-1} n_{k,m} n_{q,m+(k-q)D}^* e^{j2\pi m(l_q-l_k)/M} e^{j2\pi(k-q)Dl_q/M} \right] e^{j(\Psi_{q,l_q}-\Psi_{k,l_k})} \right\} \\
&= \frac{1}{K^2} \left\{ KP_n + \frac{1}{M^2} \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} \left[\sum_{m=0}^{M-|k-q|D-1} E \left[n_{k,m} n_{q,m+(k-q)D}^* \right] e^{j2\pi m(l_q-l_k)/M} \right] e^{j2\pi(k-q)Dl_q/M} e^{j(\Psi_{q,l_q}-\Psi_{k,l_k})} \right\} \\
&= \frac{1}{K^2} \left\{ KP_n + \frac{P_n}{M} \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} \frac{1 - e^{j2\pi(l_q-l_k)(M-|k-q|D)/M}}{1 - e^{j2\pi(l_q-l_k)/M}} e^{j2\pi(k-q)Dl_q/M} e^{j(\Psi_{q,l_q}-\Psi_{k,l_k})} \right\} \\
&= \frac{P_n}{K} + \frac{P_n}{K^2} \sum_{\substack{k=0,q=0 \\ k \neq q}}^{K-1} \frac{\sin[\pi(l_q-l_k)(M-|k-q|D)/M]}{M \sin[\pi(l_q-l_k)/M]} e^{j\pi(M-|k-q|D-1)(l_q-l_k)/M} e^{j2\pi(k-q)Dl_q/M} e^{j(\Psi_{q,l_q}-\Psi_{k,l_k})}
\end{aligned} \tag{39}$$

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